WHAT IS BETTER TO IMPROVE YOUR SPRINT?

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GARMIN

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In a previous post we saw that heavy gym strength training improve W/Kg performance for effort lasting between 1s and 40min and also durability. So, heavy gym strength training is likely to positive impact the sprint ability. The widespread principle of training specificity states that closer the training routine to the desired performance outcome, better the results. In this scenario, sprint training should be the best method to improve sprint capacity.

When the goal is to improve sprint performance (efforts lasting <30 sec), what is better: on the bike sprint training or heavy gym strength training?

A study published by Kristoffersen and Colleagues (Western Norway University of Applied Sciences, Bergen, Norway) in 2019 on Frontiers in Physiology tried to answer this question. (1)

HOW DID THEY TRAIN?

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32 (28 men, 4 women) trained cyclists (age 30, VO2max 60) were divided into two groups which for 6 weeks trained similarly (14 hrs per week). The only difference between the two groups was the following:

- **Sprint Training Group**: performed 2x week 12 x 4-8 sec all-out sprints at 110-120 rpm.
- **Gym Strength Training Group**: performed 2xweek a heavy gym strength training program consisting in 3x4-10 RM of legs exercise (half-squat, leg press, step-up, hip flexion, toe raise)

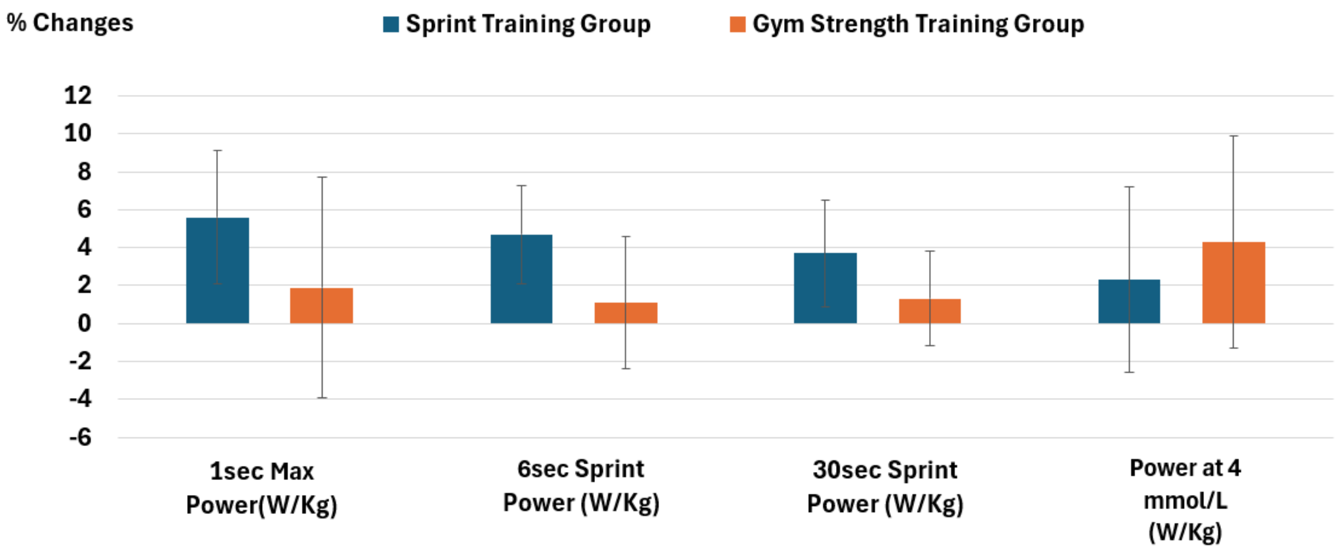
  *RM means “repetitions maximum”. So, for example, 10RM means doing 10 reps using a weight that permits to do maximum 10 repetitions (so, no repetitions in reserve). If you want to have a more detailed overview of the gym strength training program you can read more in the ‘**How Did They Train**’ section of this article.

WHAT DID THEY FIND?

- **Sprint performance improved more in the Sprint Training Group compared to the Gym Strength Training Group.**
  - 1 sec Maximal power: +5.6 ± 3.5 vs +1.9 ± 5.8 %
  - 6 sec sprint power: +4.7 ± 2.6 vs +1.1 ± 3.5 %
  - 30 sec sprint power: +3.7 ± 2.8 vs +1.3 ± 2.5 %
- **On the other hand, adaptations in power output at 4 mmol/L of blood lactate concentration tended to favour the Gym Strength Group:**

  Power at 4 mmol/L (w/kg): +2.3 ± 4.9 vs +4.3 ± 5.6 %

You can see as the % changes are reported as W/Kg. However, given that during the 6 weeks intervention body mass didn't change in both groups, the same % changes can be considered also for absolute power (Watts).



Data from Kristoffersen et al. (1).

PRACTICAL TAKEAWAYS:

> **When the goal is to improve sprint performance, Sprint training (alone) is superior to heavy gym strength training (alone)**

GABRIELE'S COMMENT

In this case the widespread principle of training specificity (closer the training routine to the performance outcome desired, better the results) seems to be confirmed: **when the goal is to improve sprint performance, Sprint training (alone) is superior to heavy gym strength training (alone)**

I think that a well designed training program to improve the sprint capacity should include both strategies: sprint training and gym strength training. However, for me this study seems to suggest tha**t if you are in a situation where you have to decide between one of the two** (for time limits, because the overall weekly load and intensity is already high and you don't have margin to include both strength and sprint training - for example during the competition season when you race on the weekend and do high intensity aerobic/anaerobic workouts during the week) **you should pick sprint training when the goal is optimizing sprint capacity,**

Thanks for Reading! Share with your Team!

Gabriele Gallo, PhD in Exercise and Sport Science

REFERENCES:

1. Kristoffersen M, Sandbakk Ø, Rønnestad BR, Gundersen H. Comparison of Short-Sprint and Heavy Strength Training on Cycling Performance. Front Physiol. 2019 Aug 28;10:1132.